Student Record and Academic Management System

Name: Walke Nikhil Bajarang

PRN: 2126UDSM1036

Department: Artificial Intelligence & Data Science (AIDS)

University: Sanjivani University

Division: A

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Problem Statement

Maintaining student academic information manually can be difficult, especially when the number of students increases. Details such as student profiles, subject-wise marks, attendance, total scores, percentages, and grades are often stored in registers, spreadsheets, or multiple files. This makes it challenging to quickly find records, update information, and keep data accurate.

Manual calculations of total marks, averages, percentages, and grades can also result in mistakes and consume valuable time. Records may be duplicated, lost, or updated incorrectly, making it difficult for teachers or administrators to track students' academic performance effectively.

To overcome these challenges, the proposed **Student Record and Academic Management System** will provide a structured and efficient way to manage student information and academic records. The system will support adding, searching, updating, and viewing student details, storing subject-wise marks, and automatically calculating academic results.

The main objective of this project is to analyse the limitations of manual student record management and design a simple, reliable, and user-friendly software solution that improves record organization, reduces errors, and makes academic information easier to manage and access.

Key Requirements

In short: The system will provide basic student record management, searching, information display, and automatic academic performance calculations.

Feature	Purpose
Add Student	Add a new student's personal information and subject marks.
Search Student	Find a student using their Roll Number or PRN.
View Student Details	Display complete student information, including academic marks.
Calculate Average Marks	Calculate the student's average marks from all subject marks.
Calculate Grade	Determine the student's grade based on their average marks.
Exit	Safely exit the system.

Inputs and Outputs

Note: The Student Record and Academic Management System accepts student information such as Roll Number, PRN, Name, Class, Division, subjects, and subject-wise marks. The system stores the information and allows users to search and manage student records. It automatically calculates total marks, average marks, percentage, and grade according to predefined grading criteria. The system provides appropriate results such as student information, calculated academic performance, or a “Student Not Found” message.

Input	Process	Output
Student record	Store student information in the system	Student record / Confirmation
Roll Number / PRN	Search for the required student record	Student details / Student Not Found
Student details / Student Not Found	Calculate total marks	Total marks
Subject-wise marks	Updated student details	Average / Percentage
Average / Percentage	Apply predefined grading rules	Grade
Updated student details	Apply predefined grading rules	Updated student record / Confirmation

Algorithm

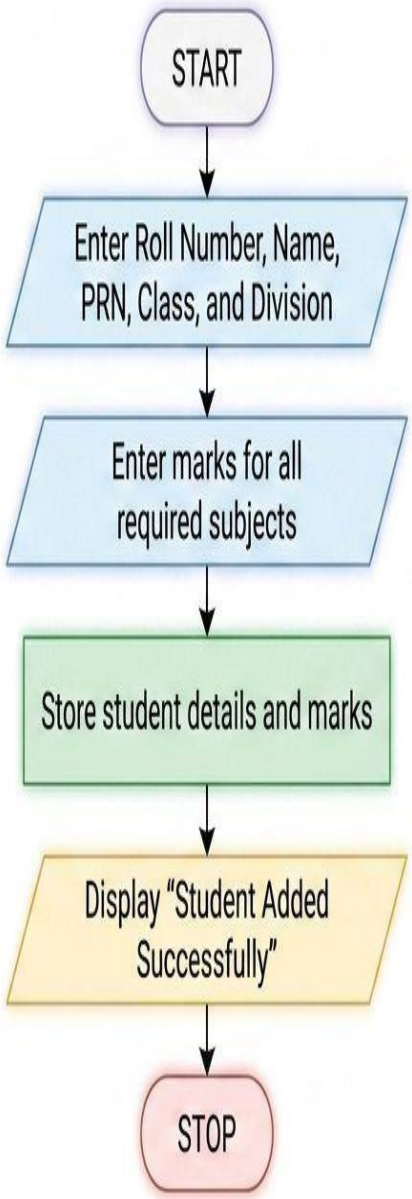
Algorithm 1 – Add Student Record 1.Start.
2.Enter Roll Number, Name, PRN, Class, and Division.
3.Enter marks for all required subjects.
4.Store the student details and marks.
5.Display "Student Added Successfully".
6.Stop.

Algorithm 2 – Search Student Record 1.Start.
2.Enter the student's Roll Number.
3.Search the student records.
4.Check whether the Roll Number exists.
4.If found, display the student's details.
5.Otherwise, display "Student Not Found".
6.Stop.

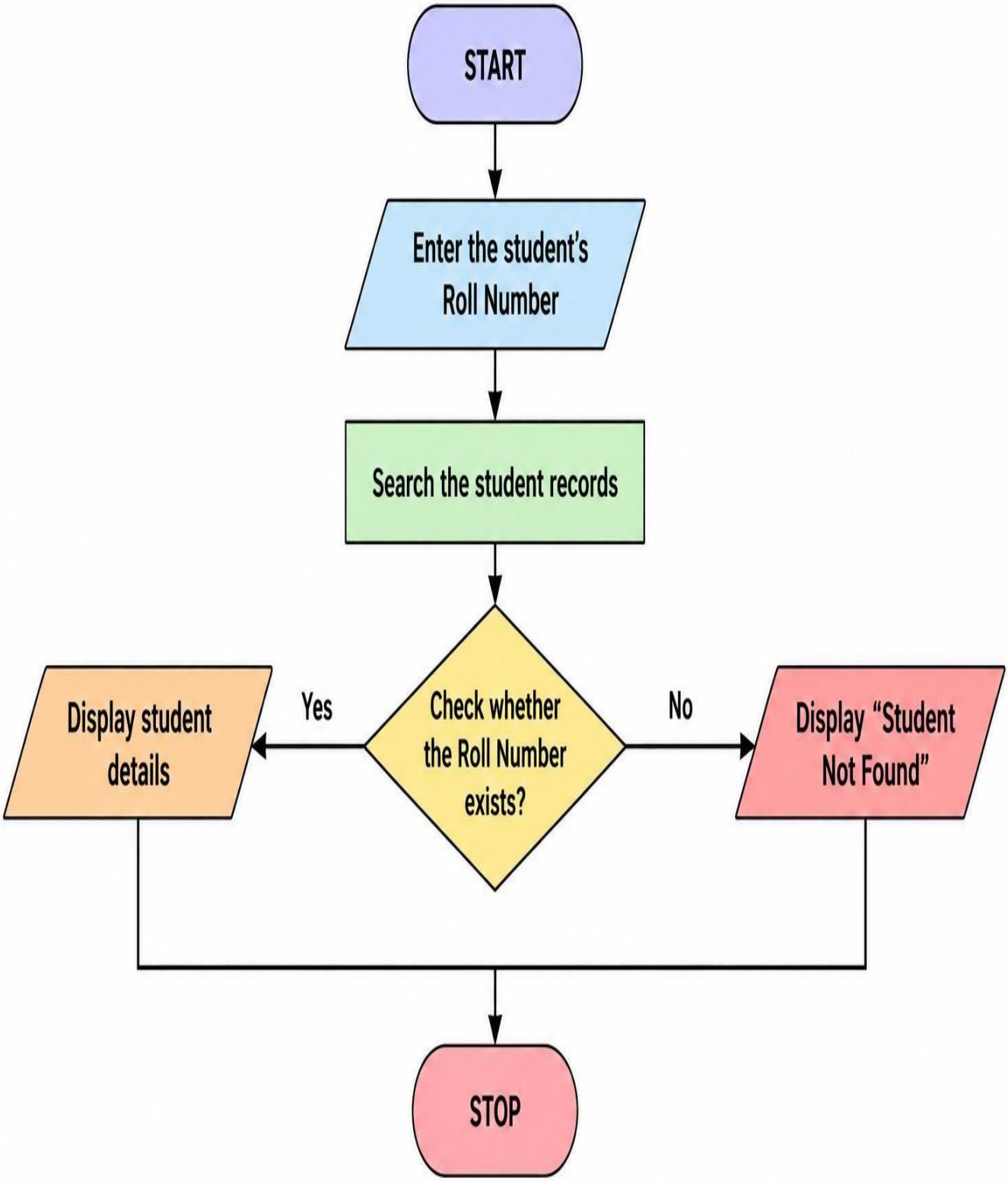
Algorithm 3 – Calculate Academic Result 1.Start.
2.Enter marks for all required subjects.
3.Calculate Total Marks.
4.Calculate Average Marks.
5.Compare the average with the grading rules.
6.Assign the appropriate Grade.
7.Display Total Marks, Average Marks, and Grade.
8.Stop.

Flowchart Collection

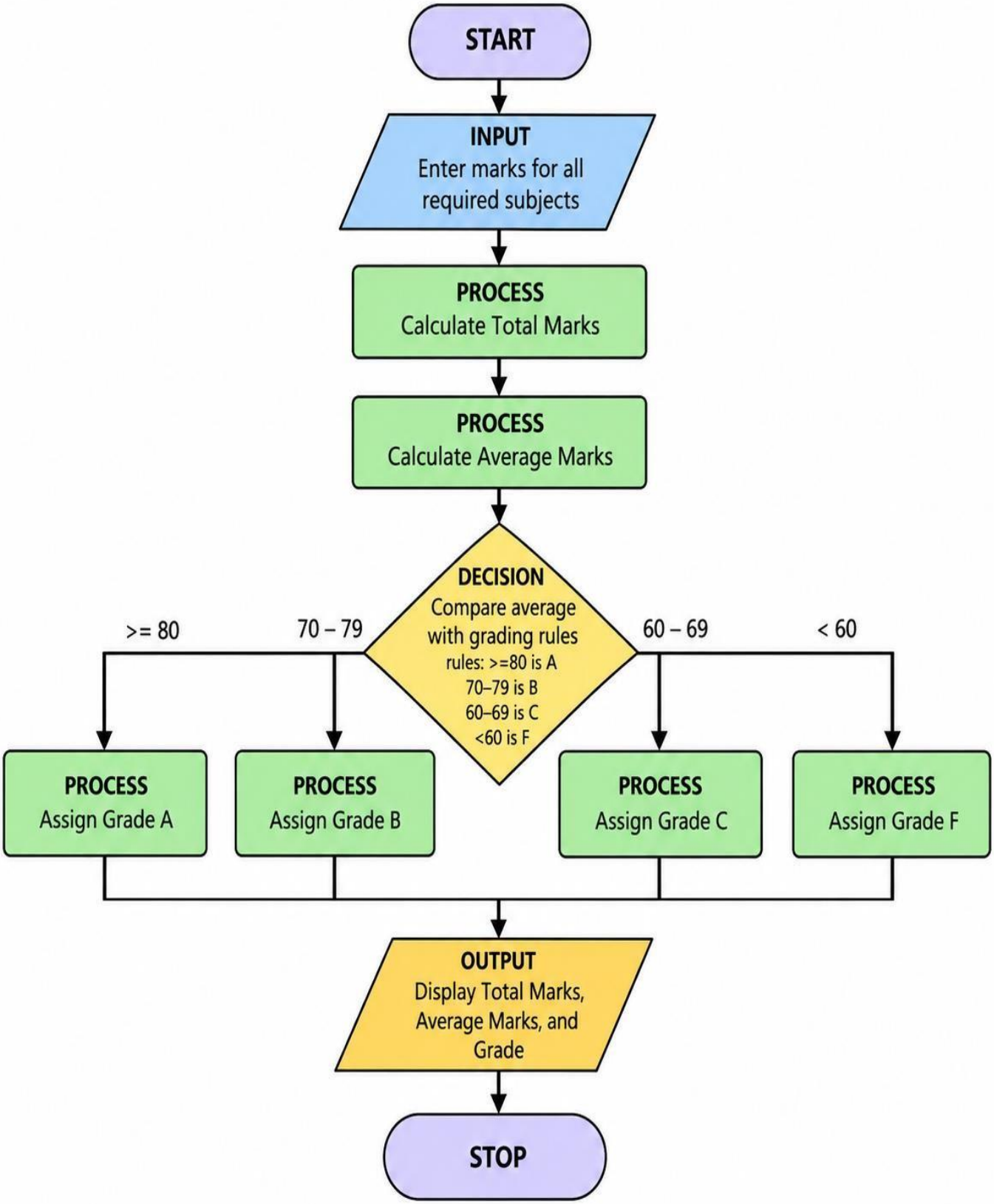
Algorithm 1 – Add Student Record



Algorithm 2 – Search Student Record



Algorithm 3 – Calculate Academic Result



Pseudocode Document

1.Pseudocode – Add Student Record

```
BEGIN

    INPUT Roll_Number

    SEARCH student records using Roll_Number

    IF Roll_Number is found THEN
        DISPLAY student details
    ELSE
        DISPLAY "Student Not Found"
    END IF

END
```

2.Pseudocode – Search Student Record

```
BEGIN

    INPUT Roll_Number

    SEARCH student records using Roll_Number

    IF Roll_Number is found THEN
        DISPLAY student details
    ELSE
        DISPLAY "Student Not Found"
    END IF

END
```

3.Pseudocode – Calculate Academic Result

```
BEGIN

    INPUT marks for all required subjects

    CALCULATE Total_Marks
    CALCULATE Average_Marks

    IF Average_Marks >= 80 THEN
        Grade = "A"
    ELSE IF Average_Marks >= 70 THEN
        Grade = "B"
    ELSE IF Average_Marks >= 60 THEN
        Grade = "C"
    ELSE
        Grade = "F"
    END IF

    DISPLAY Total_Marks
    DISPLAY Average_Marks
    DISPLAY Grade

END
```

Short Explanation of Proposed Solution

The proposed Student Record and Academic Management System is designed to provide a simple and organized way to manage student information and academic records. It will store important details such as Roll Number, Name, PRN, Class, Division, and subject-wise marks in one system.

The system will provide different operations such as Add Student, Search Student, View Student Details, Calculate Average Marks, and Calculate Grade. When adding a student, the user will enter the required personal and academic information, which will then be stored for future use.

The Search Student feature will allow the user to find a particular student's record using the Roll Number. If the record is available, the system will display the student's details; otherwise, it will show "Student Not Found."

The system will also process the marks entered for each subject. It will calculate the total marks and average marks and determine the student's grade according to the approved grading criteria. This reduces manual calculations and helps avoid common calculation errors.

Overall, the proposed system will make student record management faster, easier, organized, and more reliable. It can also provide a foundation for adding more features in the future, such as updating records, deleting records, generating reports, and storing data in a database.